Roan Mountain, State Hwy 143 (Carter 099 046.02)

13.29 ac · Graded

Graded on an earlier boundary: the parcel's lines changed after this grade. Grade it again for a current report card.



Filled: the best site · Imagery: USGS (public domain)

B **Maybe** · Profile 1: Strong elevation and winter sun; check first: driveway on your land and septic. Held at Maybe by: Driveway on your land and Septic.

MUST HAVE

Floodplain	✓	No FEMA flood zone on the parcel
Legal road access	✓	Frontage on an unnamed road
Driveway on your land	!	Check first: the driveway needs an easement across neighbouring land
Septic	!	Check first: no soil on the parcel rates better than very limited for a septic field

REALLY WANT

Elevation at the build site	B	2,710 ft at site #1
Best homesite	A	faces S, 7.3 h of December sun, 8° slope
Room to build	D	0.5 ac of gentle ground near the house site, 1.2× the build footprint
Night sky	C	rural–suburban sky (zone 4a)
Trails and public land	A	a trailhead 3 min (1.0 mi) by road, 0.68 mi straight (Roan Mountain State Park); serves: Peg Leg Mine Trail
Garden ground	B	the largest garden is 0.34 ac
Town and hospital	C	a 32-minute drive to an ER, 5 to a grocery

QUESTIONS FOR THE AGENT

1. Does the driveway need to cross neighbouring land? If so, is there a recorded easement for it?
2. Has the land had a perc test or a soil evaluation? Is a conventional septic system possible?

Still unknown not in any dataset

- [ ] Legal access: deeded, recorded, width, maintenance agreement
- [ ] Deed restrictions / CC&Rs — read the actual recorded document
- [ ] Severed mineral or timber rights — check the deed chain, not the listing
- [ ] Utility extension quote from the power co-op, by parcel ID
- [ ] Neighboring well depths and yields (county health department)
- [ ] Septic: get a soil scientist's opinion on the bench, not the pin
- [ ] Whether RV or shop-house occupancy during the build is lawful (zoning office)
- [ ] Homeowner's insurance quote for the address before you fall in love
- [ ] Cell service and the Starlink sky view from the bench

THE PRIORITIES IT WAS GRADED AGAINST

Must have

- No flood zone at the house or the driveway
- Reachable by a public road, or a recorded right-of-way
- The driveway stays on your land, or its easement is recorded
- Soil that can take a septic field

Really want

1. **Elevation at the build site** (25%): 2,500 ft or higher
2. **Best homesite** (25%): A sunny, gentle spot for the house
3. **Room to build** (15%): Enough gentle ground for the house
4. **Night sky** (10%): Dark enough to see the Milky Way
5. **Trails and public land** (10%): A trailhead within 20 min, or public land within 1 mi
6. **Garden ground** (10%): Room for a kitchen garden
7. **Town and hospital** (5%): ER within 30 min, groceries within 20 min

Don't care: Driveway

BEST SITE AND ALTERNATIVES

1. **1st · Site #1**: Elevation 2,710 ft · Winter sun 7.3 h sun, faces S · Room to build 1.2× footprint · Night sky zone 4a
2. **2nd · Site #3**: Elevation 2,752 ft · Winter sun 7.6 h sun, faces S · Room to build 1.1× footprint · Night sky zone 4a
3. **3rd · Site #2**: Elevation 2,735 ft · Winter sun 6.9 h sun, faces SSW · Room to build 0.3× footprint · Night sky zone 4a
4. **4th · Site #4**: Elevation 2,920 ft · Winter sun 7.9 h sun, faces SE · Room to build house only · Night sky zone 4a

Where to build ranked from the data

1 Best site — A site — $$$$ to build

overall 67 · 2,710 ft · 36.18505, -82.07373

sun **76** aspect **100** frost **100** slope **89** sky **56** driveway **2,367 ft**

- 7.3 h of December sun — good
- 0.9 ac, faces S, 7.8° slope, 158 ft above the valley floor
- Tusquitee: septic rated very limited — alternative system
- routed driveway: 2367 ft at ≤15%, 6 switchbacks, ~$80–148k, needs an easement
- Within 15% only via neighbouring land (needs an easement): 2367 ft

2 Runner-up — B site — $$$$ to build

overall 64 · 2,735 ft · 36.18588, -82.07491

sun **72** aspect **100** frost **100** slope **78** sky **56** driveway **2,012 ft**

- 6.9 h of December sun — acceptable
- 0.3 ac, faces SSW, 9.4° slope, 183 ft above the valley floor
- Tusquitee: septic rated very limited — alternative system
- 649 ft to unnamed road at 23% straight-line — about 979 ft of driveway with switchbacks
- routed driveway: 2012 ft at ≤15%, 6 switchbacks, ~$75–140k, needs an easement
- Within 15% only via neighbouring land (needs an easement): 2012 ft

3 Third (compact) — B site — $$$$ to build

overall 60 · 2,752 ft · 36.18556, -82.07424

sun **79** aspect **100** frost **100** slope **55** sky **56** driveway **2,185 ft**

- compact site: 0.48 ac shelf — needs a cut pad and probably a retaining wall
- 7.6 h of December sun — good
- 0.5 ac, faces S, 13.3° slope, 201 ft above the valley floor
- Porters: septic rated very limited — alternative system
- 591 ft to State Rte 143 at 17% straight-line — about 663 ft of driveway with switchbacks
- routed driveway: 2185 ft at ≤15%, 6 switchbacks, ~$78–144k, needs an easement
- Within 15% only via neighbouring land (needs an easement): 2185 ft

Side by side

Site	Quality	Sun h	Aspect	Frost	Slope	Sky	Cost	Septic	Found.	Rock	Pad	Driveway	Overall
#1	**A** 82	7.3	100	100	89	56	**$$$$** 69	40	0	0	2	2,367 ft (27)	67
#2	**B** 79	6.9	100	100	78	56	**$$$$** 71	40	0	0	4	2,012 ft (27)	64
#3 compact	**B** 78	7.6	100	100	55	56	**$$$$** 84	40	0	1.9500000000000002	15	2,185 ft (27)	60
#4 compact	**A** 85	7.9	100	100	89	59	**$$$$** 78	40	0	1.9500000000000002	15	652 ft (21)	66

Quality is the spot itself — sun 40, aspect/frost/slope/sky 15 each — and can't be bought. Cost is what it takes to build there, as points toward 100: septic class, foundation rating, rock, pad cutting, driveway length and grade. A great spot with a long driveway is an A site at $$$, not a D.

Shelves for a shop, barn or garage

- **S1** — 0.48 ac at 2,752 ft, faces S, 13.3° slope, cell score 54; 238 ft from site #1, up 42 ft
- **S2** — 0.20 ac at 2,920 ft, faces SE, 7.7° slope, cell score 80; 1,041 ft from site #1, up 211 ft
- **S3** — 0.15 ac at 2,776 ft, faces SSE, 11.4° slope, cell score 65; 657 ft from site #1, up 67 ft
- **S4** — 0.14 ac at 2,742 ft, faces ESE, 11.1° slope, cell score 62; 555 ft from site #1, up 32 ft

A shelf is ground that scores 45+ but is too small or too shaded for a house — fine for a building you don't live in. Watch the drop from the house: 60 ft down over 400 ft is a 15% driveway.

Ranked by overall = 70% site quality + 30% (100 – cost). It ranks the ground the terrain scan found; it doesn't know about views, wells, or the neighbor's dog.

Terrain

Elevation	2,666–3,161 ft (mean 2,842)
Relief across the parcel	494 ft
Height above local valley floor	268 ft (floor 2,551 ft)
Slope, median / 90th pct	24.7° / 32.9°
Acres under 15% grade	0.8 of 13.3
Acres over 25% grade	10.6
House sites found	2 (1.27 ac) · shelves 4 · garden patches 2
Largest house site	0.95 ac at 2,710 ft, faces S, 7.8° slope, cell score 83
Nearest road to bench	State Rte 143: climb 41 ft over 439 ft (9% straight-line)

Of 13.3 ac: 1.9 ac score ≥60 for a house, 0.9 ac more ≥45 (shelf grade), 0.5 ac ≥60 for a garden. DEM: USGS 3DEP at 3.0 m.

ACCESS AND DRIVEWAY

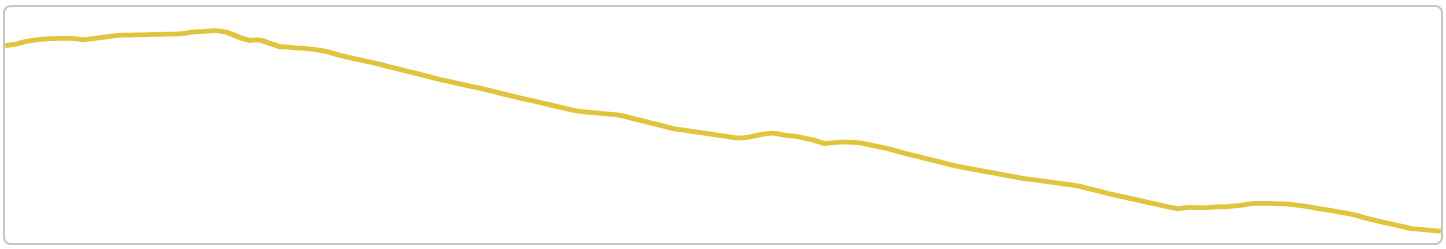
- **Legal road access:** Frontage on an unnamed road
- **Driveway on your land:** Check first: the driveway needs an easement across neighbouring land

Driveway routed to site #1

Entrance candidates on the frontage (E1 best): E1 road — road 38%, bend 12°, bank 1 ft.

Recommended — shortest legal — needs an easement from E1 **$80–148k**

Length / rise	2,367 ft / 182 ft
Grade over 30 m / average	16.0% / 10.2% (limit 15%)
Switchbacks	6
Earthwork	2,125 yd^3 (41 yd^3 in rock)
Stone	724 tons (12 ft × 6 in)
Clearing corridor	1.30 ac
Culverts	1 (entrance + 0 drainage crossings)
Relief culverts	9 (one every 150–400 ft, by grade)
Paving (over 12%)	~1,270 ft (cost TBD)
Leaves the boundary	**404 ft outside the line — needs an easement**



▼ Grade over 15 and 60 m

Grade over 15 m	18.9%
Grade over 60 m	14.7%

Grades are measured on the ground under the route, between the DEM's cells. A built road's grade reads about 1–4 points lower, once cut and fill smooth the ground.

Pioneer 4×4 track first

On the driveway's alignment (keeps the dozer work)	~25 machine hours, 13 water bars · **$11–21k**
Direct line at ≤15% (red dotted; cheaper now, replaced later)	2,330 ft, 17% over 30 m · **$12–23k**

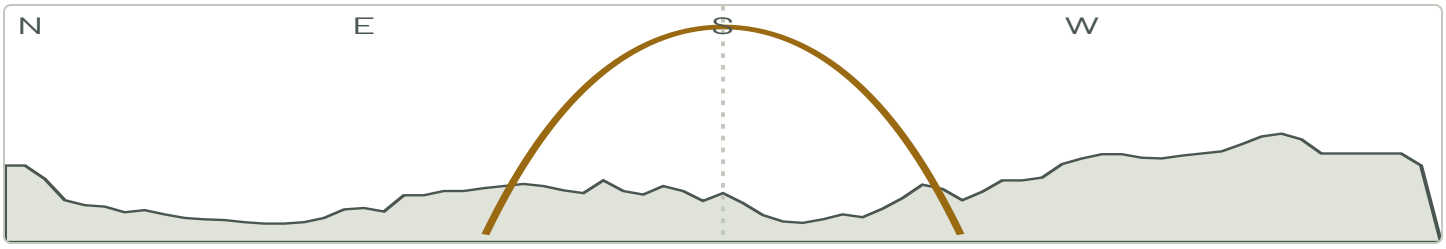
A track is blade width, native soil, water bars instead of ditches, a pipe or ford at crossings, stone only at the entrance. Building it on the final alignment means the earthwork carries forward; the direct line is money spent twice.

Least-cost route over the 3 m lidar with a hard grade limit; it avoids bottomland and side-hill, and prices crossings as culverts. Costs use the unit rates in Settings, ±30%. Bare-earth DEM (no trees), county soils (not borings), and VDOT decides the entrance — this is a number to put in front of an excavator, not a bid.

WINTER SUN AND THE HORIZON

December sun at the largest house site, 2,706 ft

This point	2,706 ft elevation, 155 ft above the valley floor, 3.0° slope facing S · 36.18505, -82.07373



Direct sun, Dec 21	7.3 of 9.6 daylight hours
Noon sun altitude / worst southern ridge	30.4° / 8.7° at 150° (SSE)
Noon clearance (after 3° canopy)	18.7°
Direct sun, Jun 21	12.6 of 14.5 h

Grey is bare-earth terrain from lidar; trees add to it.

NIGHT SKY

Dark skies score 56/100 at the largest house site

Sky at zenith	20.92 mag/arcsec^2 — zone 4a, rural–suburban
Artificial vs natural light	1.70× (1.0 = half the sky's light is man-made)
Milky Way core, peak altitude	24.8° above due south
Southern ridge / clearance	8.7° / 13.2° (after 3° canopy)
Brightest patch to the south	2.74× at 20 mi, bearing 180° (S)

- The core peaks 25° above due south; the ridge there is 9° — 13° of clear sky under it.
- Bright patch toward S at 20 mi (ratio 2.7): expect a light dome low in the S exactly where the core sits.

Zenith brightness from the Light Pollution Atlas 2025 (Lorenz, after Falchi/Cinzano), 1/120° grid. The atlas is zenith-only, so the southern-dome line samples the ground map toward the core as a proxy. The core is up in the evening from June to September.

SOILS, SEPTIC AND FLOOD

Soils what the ground will let you do

NRCS mapped the county into units. Each unit is a mix of soils they expect you'd find if you dug. Slope in the unit name tells you which part of the parcel it is.

Tusquitee loam, 20 to 35 percent slopes 5.8 ac inside the line

Tusquitee 100% of this unit

House & septic — Too steep for a house or drainfield without heavy earthwork.

Garden & animals — Woodlot. Goats will browse it; nothing else belongs there.

Well drained; septic "Very limited"; house w/ basement "Very limited"; no basement "Very limited"; roads "Very limited"; flooding None; bedrock not recorded; hydric No

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Porters gravelly loam, 50 to 80 percent slopes, stony 4.1 ac inside the line

Porters 100% of this unit

House & septic — Too steep for a house or drainfield without heavy earthwork.

Garden & animals — Woodlot. Goats will browse it; nothing else belongs there.

Well drained; septic "Very limited"; house w/ basement "Very limited"; no basement "Very limited"; roads "Very limited"; flooding None; bedrock 137 cm; hydric No

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Porters gravelly loam, 30 to 50 percent slopes, stony 3.4 ac inside the line

Porters 100% of this unit

House & septic — Too steep for a house or drainfield without heavy earthwork.

Garden & animals — Woodlot. Goats will browse it; nothing else belongs there.

Well drained; septic "Very limited"; house w/ basement "Very limited"; no basement "Very limited"; roads "Very limited"; flooding None; bedrock 137 cm; hydric No

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Map-unit lines are drawn at county scale, so a boundary can be 100 ft off on the ground.

Floodplain

FEMA has mapped this area; nothing on the parcel is in a Special Flood Hazard Area (X).

Where to garden frost-safe, sunny, gentle, good dirt

G1 **0.34 ac** A 87/100 · 2,706 ft, 155 ft above the valley floor · faces S, 3.7° · well drained (Tusquitee)

G2 **0.06 ac** A 84/100 · 2,922 ft, 370 ft above the valley floor · faces SE, 4.1° · well drained (Porters)

Garden score weighs frost position heavily: cold air drains downhill on still clear nights and pools in the low ground, where the last spring frost comes weeks later and the first fall frost weeks earlier. Slope ≤6° and a southern face do the rest; soil drainage and NRCS farmland class adjust the score.

DRIVE TIMES

Getting there and getting out

Nearest hospital Cannon Memorial Hospital	32 min, 18.2 mi
Nearest real grocery Ingles	23 min, 13.3 mi Closer: RediMart Supermarket, 5 min.
Asheville airport (AVL)	118 min, 86.4 mi
Roanoke airport (ROA)	228 min, 188.6 mi
Tri-Cities airport (TRI)	64 min, 41.8 mi
Greenville-Spartanburg (GSP)	183 min, 124.6 mi
Charlotte airport (CLT)	179 min, 119.9 mi
Groceries within 28 mi (straight line)	RediMart Supermarket 1 mi RediMart Supermarket 9 mi Thomas Grocery 9 mi Ingles 10 mi Just Local Market 13 mi Food Lion 13 mi
Trailheads within 22 mi	124 (nearest 0.7 mi)

Drive times from OSRM's public router — fine for comparing parcels, not for catching a flight. Places come from OpenStreetMap data (via Photon); trailhead counts undercount national forest access. Roads from the Census Bureau's TIGER lines. Forest Service trailheads and state parks are included too. Drive times start at the driveway's entrance on an unnamed road.

Public land within a mile

- **Cherokee National Forest** (U.S. Forest Service, National Forest) — 4,939 ft away; access open

Conservation easements on private land are not in this layer and don't count — you can't walk on them.

LISTING FACTS AND NOTES

No listing facts yet.

LINES AND MARKS

None drawn.

DATA SOURCES

- USGS 3DEP lidar (USGS 3DEP, 3 m): as of the screen
- NRCS soils (Soil Data Access): as of the screen
- FEMA flood map (NFHL): as of the screen
- USGS PAD-US (public land): as of the screen
- Light pollution atlas (D. Lorenz, 2025)
- Census TIGER roads: as of the screen
- OpenStreetMap places and trailheads: as of the screen
- OSRM drive times: as of the screen

Graded by engine version 22.

This is a desk screen, not a survey, perc test or appraisal.